

mPhilGEPS Production BOM

Production-Grade Bill of Materials (BOM) & 3-Year Lifecycle Cost Analysis

Modernized Philippine Government Electronic Procurement System (mPhilGEPS) Phase 2

Target Cloud Platform: Google Cloud Platform, Gemini Enterprise & Google Sovereign AI Stack
Primary Region: Singapore (`asia-southeast1`) | **Disaster Recovery Region:** Jakarta (`asia-southeast2`)
Statutory Governance: RA 12009 (New Government Procurement Act / NGPA) & RA 10173 (Data Privacy Act)
Currency Benchmark: USD (\$) and Philippine Peso (PHP ₱) at official budgetary baseline rate: **\$1.00 USD = ₱65.00 PHP**

1. Executive Summary & Cost Architecture

This production-grade Bill of Materials (BOM) provides the comprehensive financial model, operational sizing, and business continuity architecture for the **Modernized Philippine Government Electronic Procurement System (mPhilGEPS Phase 2)**, in direct response to the requirements outlined in the Terms of Reference (TOR), the Business Requirements Document (BRD), and the Technical Design Document (TDD).

Strategic Architecture Upgrade: BigQuery Enterprise Edition Slots (Predictable vs. On-Demand)

To eliminate catastrophic “bill shock” from unpartitioned ad-hoc scans or runaway analytical queries by observers and auditors, the BigQuery architecture is transitioned from unpredictable On-Demand per-TB pricing (\$6.88/TB scanned) to **BigQuery Enterprise Edition Autoscaling Slots (25 Baseline, 100 Max Cap)**.

This architectural upgrade delivers four decisive advantages: 1. **Guaranteed Budgetary Ceiling:** Compute capacity is capped at 100 slots within a dedicated reservation (`mphilgeps-analytics-reservation`). Queries execute within governed capacity and **can never exceed the budget**. 2. **Zero-Cost In-Database Machine Learning:** Under Enterprise Edition, **BigQuery ML (`ARIMA_PLUS` & K-means clustering)** uses the provisioned reservation slots. The expensive **\$312.50/TB** specialized ML training charge is **completely eliminated (\$0.00)**. 3. **Enterprise Governance & Security:** Unlocks native **VPC Service Controls**, **Customer-Managed Encryption Keys (CMEK)** via Cloud KMS HSM, and column/row-level security required by **RA 10173**. 4. **Commercial Anti-Scraping & Open Data De-Monetization:** Integrates **reCAPTCHA Enterprise** (500,000 frictionless behavioral risk assessments/mo enforced at the Cloud Armor edge) with **JA3 TLS fingerprinting** and **Dedicated Open Contracting Data Standard (OCDS) Bulk Exports** on Cloud Storage/CDN—protecting shared government agency NAT IP gateways while eliminating the commercial moat of paid third-party data scrapers.

Financial Synthesis

Cost Dimension	On-Demand (List Price)	1-Year Committed Use (CUD)	3-Year Committed Use (CUD)
Monthly Recurring Cost (USD)	\$22,588.66	\$17,709.05	\$14,586.55
Monthly Recurring Cost (PHP @ ₱65)	₱1,468,262.90	₱1,151,088.25	₱948,125.75
Annualized Spend (USD)	\$271,063.92	\$212,508.60	\$175,038.60

Cost Dimension	On-Demand (List Price)	1-Year Committed Use (CUD)	3-Year Committed Use (CUD)
Annualized Spend (PHP @ ₱65)	₱17,619,154.80	₱13,813,059.00	₱11,377,509.00
3-Year Lifecycle Total (USD)	\$853,500.00	\$669,500.00	\$551,808.60
3-Year Lifecycle Total (PHP @ ₱65)	₱55,477,500.00	₱43,517,500.00	₱35,867,559.00
Effective Cost Reduction	Baseline	-21.6%	-35.4%

+-----+ + MONTHLY COST DISTRIBUTION (USD) [Category 1] Compute & Containers (GKE, Cloud Run, Confidential Space) : \$3,186.00 (14.1%) [Category 2] Databases (AlloyDB HA, Spanner Graph, Redis Cluster) : \$3,102.80 (13.7%) [Category 3] Data Warehouse & Governed Slots (BQ Slots, BQML, Looker) : \$6,480.38 (28.7%) [Category 4] AI & Multi-Agent Hub (Gemini 3.x, DocAI, Vector Search) : \$702.58 (3.1%) [Category 5] Perimeter, Network, API & Bot Defense (Armor, reCAPTCHA) : \$5,362.10 (23.7%) [Category 6] Sovereign Security & SOC (Cloud KMS HSM, SecOps Chronicle) : \$931.50 (4.1%) [Category 7] Storage & 10-Yr WORM Compliance (GCS Standard & Archive) : \$298.30 (1.3%) [Category 8] Observability & Support (Cloud Ops, Gemini Ent, 24/7 SLA) : \$2,525.00 (11.3%) TOTAL MONTHLY RUNTIME (GOVERNED SLOTS & ANTI-SCRAPING LIST PRICE) : \$22,588.66 (100.0%) +-----+ +			
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2. Workload Sizing, User Concurrency & Traffic Spike Analysis

2.1 User Base Breakdown: Registered vs. Active vs. Concurrent

In mission-critical enterprise systems, infrastructure sizing is determined strictly by **Peak Concurrent Users (PCU)** and **Requests Per Second (RPS)**, rather than total account registrations.

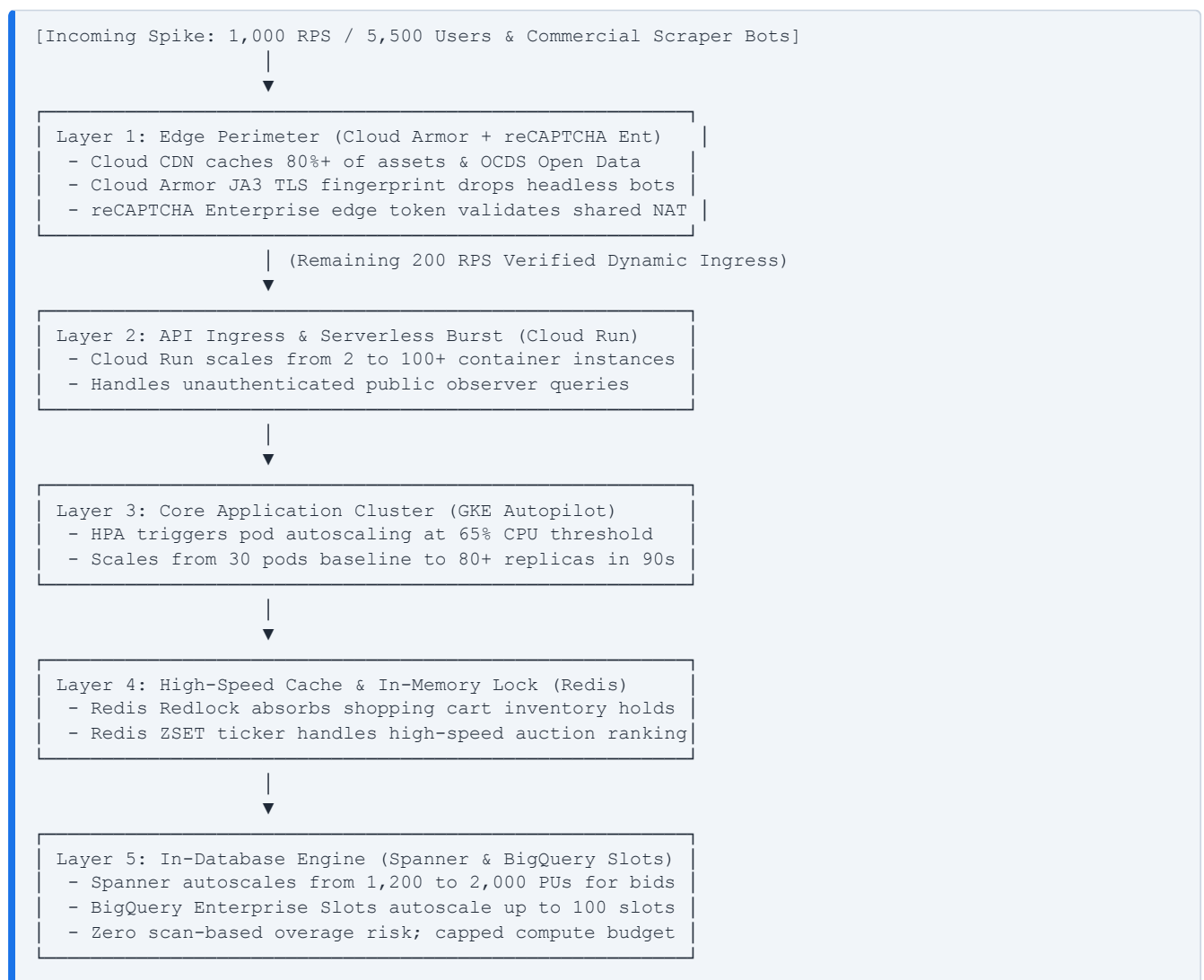
+-----+ -+ USER DEMOGRAPHY & LOAD MODEL User Tier Scale Metric Activity Profile ----- Total Registered Entities 5,000+ Procuring Entities Government agencies, LGUs, SUCs, GOCCs Total Registered Vendors 10,000 to 100,000+ Merchants in GOP-OMR (Red & Platinum) Daily Active Users (DAU) 15,000 to 25,000 / day Procurement officers, bidders, BAC secretariats Baseline Concurrent Users 3,500 active sessions Regular business hours portal transactions Peak Surge Concurrency 5,500+ concurrent users Month-end deadlines & bid-closing windows +-----+ -+			
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2.2 Throughput & Requests-Per-Second (RPS) Translation

- **Baseline Load (3,500 Concurrent Users):**
 - Average user interaction interval: 15–20 seconds between page navigations, searches, and catalog clicks.
 - Translates to **150 to 250 HTTP Requests Per Second (RPS)** reaching the application layer.
- **Peak Surge Load (5,500 Concurrent Users):**
 - During the final 30 minutes before bid submission deadlines (typically 9:00 AM – 12:00 PM), interaction frequency jumps to 5–8 seconds per user due to status polling, WebSocket clock refreshes, and multi-megabyte document uploads.
 - Translates to **600 to 1,000 HTTP Requests Per Second (RPS)** at the edge.

2.3 Multi-Layer Spike Absorption Architecture

To ensure sub-second response times (< 1,000 ms) without paying for idle over-provisioned VMs, traffic spikes are absorbed sequentially across five layers:



3. Categorized Production Bill of Materials (Singapore: asia-southeast1)

Category 1: Core Compute, Container Platform & Confidential Enclaves

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
1.1	GKE Autopilot (vCPU)	55 vCPU monthly average across 10 domain microservices (HPA autoscaling up to 80 vCPU)	\$0.0535 / vCPU-hr	\$2,148.03	₱139,621.95	\$25,776.36
1.2	GKE Autopilot (Memory)	150 GiB RAM monthly average across pods	\$0.0058 / GiB-hr	\$635.10	₱41,281.50	\$7,621.20
1.3	GKE Autopilot (Storage)	120 GiB Ephemeral Pod Storage	\$0.00006 / GiB-hr	\$5.26	₱341.90	\$63.12
1.4	GKE Cluster Fee	Multi-Zonal Management Fee (covered by \$73/mo default cluster credit)	\$0.10 / hr	\$0.00	₱0.00	\$0.00
1.5	Cloud Run (Serverless)	Public chat widget & unauthenticated observer endpoints (2.5M requests/mo, 250ms avg, 2 warm instances)	Serverless invocations + CPU/RAM allocation	\$22.39	₱1,455.35	\$268.68
1.6	Confidential Space Bid Vault	2 x n2d-standard-4 Confidential VMs (8 vCPU, 32 GB RAM total) with AMD SEV-SNP hardware memory encryption	\$0.257 / VM-hr (incl. Confidential surcharge)	\$375.22	₱24,389.30	\$4,502.64
SUB	Category 1 Subtotal	Core Compute & Container Platform	—	\$3,186.00	₱207,090.00	\$38,232.00

Category 2: Relational, Graph & Distributed In-Memory Databases

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
2.1	Cloud Spanner (Compute)	1,200 Processing Units (1.2 Nodes equivalent) in asia-southeast1 . Houses contract ledger & Spanner Graph	\$0.0009 / PU-hr (\$0.90/1000 PU-hr)	\$788.40	₱51,246.00	\$9,460.80
2.2	Cloud Spanner (Storage)	500 GB Regional SSD Storage	\$0.30 / GB-mo	\$150.00	₱9,750.00	\$1,800.00
2.3	Cloud Spanner (Backups)	1,500 GB automated daily incremental backups (30-day retention)	\$0.056 / GB-mo	\$84.00	₱5,460.00	\$1,008.00
2.4	AlloyDB Enterprise (Primary)	8 vCPU, 64 GB RAM HA Instance with multi-zone standby failover. (AlloyDB HA compute for standby is zero-charge)	\$0.075/vCPU-hr + \$0.0125/GB-hr	\$1,022.00	₱66,430.00	\$12,264.00
2.5	AlloyDB Read Pool Instance	1 x Read Pool Instance (4 vCPU, 32 GB RAM) for eMarketplace catalog lookups & merchant queries	\$0.075/vCPU-hr + \$0.0125/GB-hr	\$511.00	₱33,215.00	\$6,132.00
2.6	AlloyDB Storage & Backups	1,000 GB autoscaling SSD storage + 1,500 GB continuous Point-In-Time Recovery (PITR) snapshots	\$0.18/GB-mo storage + \$0.06/GB-mo backup	\$270.00	₱17,550.00	\$3,240.00

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
2.7	Memorystore for Redis (HA)	16 GB Standard Tier HA instance with cross-zone replication. Handles Redlock cart holds & ZSET auction ticker	\$0.38 / hr	\$277.40	₱18,031.00	\$3,328.80
SUB	Category 2 Subtotal	Relational, Graph & Distributed Databases	—	\$3,102.80	₱201,682.00	\$37,233.60

Category 3: Enterprise Data Warehouse, Governed Slots & In-Database AI

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
3.1	BigQuery Active Storage	3,000 GB (3 TB) operational analytical tables and staging data	\$0.020 / GB-mo	\$60.00	₱3,900.00	\$720.00
3.2	BigQuery Long-Term Storage	10,000 GB (10 TB) unmodified historical procurement logs (>90 days old)	\$0.010 / GB-mo	\$100.00	₱6,500.00	\$1,200.00
3.3	BigQuery Streaming Ingestion	Storage Write API ingesting ~10M state transitions/audit events/mo	\$0.025 / GB	\$0.38	₱24.70	\$4.56
3.4	BigQuery Enterprise Slots	Governed Capacity Reservation: 25 Baseline Slots autoscaling up to 100 Max Slots (~20,000 slot-hours/mo). Zero on-demand scan cost risk.	\$0.066 / slot-hr	\$1,320.00	₱85,800.00	\$15,840.00

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
3.5	BigQuery ML (ARIMA_PLUS)	Included in Slot Reservation: Weekly demand forecasting retraining (holiday_region = 'PH') & CPI clustering. Billed against slots at \$0 additional per-TB cost.	\$0.00 / TB (Covered by Slots)	\$0.00	₱0.00	\$0.00
3.6	Looker Core (Enterprise BI)	Looker Core instance with embedded analytics for COA observers, public transparency, and PS-DBM dashboards	\$5,000.00 / mo	\$5,000.00	₱325,000.00	\$60,000.00
SUB	Category 3 Subtotal	Data Warehouse, Governed Slots & In-DB AI	—	\$6,480.38	₱421,224.70	\$77,764.56

Note on BI Alternatives: If PS-DBM elects to deploy **Looker Studio Pro** with 30 named creator licenses (\$9/user/mo = \$270.00/mo) in lieu of Looker Core, Category 3 subtotal drops to **\$1,750.38 / month** (₱113,774.70 PHP), reducing the total monthly bill by \$4,730.00 USD.

Category 4: Sovereign AI & Multi-Agent Ecosystem

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
4.1	DocAI Custom Extractor	7,500 pages/mo (SEC GIS, DTI, PCAB Construction Licenses) for GOP-OMR accreditation	\$50.00 / 1,000 pages	\$375.00	₱24,375.00	\$4,500.00
4.2	DocAI Form Parser	2,500 pages/mo (BIR Tax Clearance Certificates with barcode validation)	\$30.00 / 1,000 pages	\$75.00	₱4,875.00	\$900.00
4.3	DocAI Layout Parser	2,500 pages/mo (Audited Financial Statements for NFCC computation)	\$30.00 / 1,000 pages	\$75.00	₱4,875.00	\$900.00

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
4.4	Gemini 3.1 Pro (Reasoning)	15M input tokens (\$1.25/M) + 5M output tokens (\$5.00/M) for UNSPSC Pass 2 JSON Schema enforcement & AFS audit	Tiered Token Pricing	\$43.75	₱2,843.75	\$525.00
4.5	Gemini 3.7 Flash (Fast)	120M input tokens (\$0.075/M) + 30M output tokens (\$0.30/M) for Pass 1 normalization & Supervisor routing	Tiered Token Pricing	\$18.00	₱1,170.00	\$216.00
4.6	Vertex AI Vector Search	text-embedding-005 (5M chars/mo) + 1 x Standard Index Endpoint for proactive ITB vendor-tender matching	\$0.025/M chars + \$0.09/endpoint-hr	\$65.83	₱4,278.95	\$789.96
4.7	Google Cloud Model Armor	1,000,000 prompt/response inspections/mo for PII redaction (TIN, phone numbers) & prompt injection defense	\$0.50 / 10,000 inspections	\$50.00	₱3,250.00	\$600.00
SUB	Category 4 Subtotal	Sovereign AI & Multi-Agent Ecosystem	—	\$702.58	₱45,667.70	\$8,430.96

Category 5: Edge Perimeter, Networking, API Management & Anti-Scraping Bot Defense

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
5.1	Apigee API Management	1 Production Environment (\$0.20/hr) + 4,000,000 API calls/mo for external integrations (BTMS, SEC, BIR, Banks)	Hourly env + Tiered API call fee	\$796.00	₱51,740.00	\$9,552.00
5.2	Cloud Armor Enterprise	Managed Protection Plus: 24/7 DDoS mitigation, JA3 TLS fingerprinting, OWASP Top 10, endpoint rate limiting	\$3,000.00 / mo	\$3,000.00	₱195,000.00	\$36,000.00
5.3	Global HTTPS Load Balancer	Global External Application Load Balancer with Google-managed SSL & HTTP/3 (10 TB data processed/mo)	\$0.025/hr + \$0.008/GB	\$98.25	₱6,386.25	\$1,179.00
5.4	Cloud CDN (Edge & Open Data)	Edge caching for public portal static assets, catalogs, tender PDFs, and OCDS Open Data Bulk Feeds (5 TB cached egress/mo)	\$0.050 / GB	\$250.00	₱16,250.00	\$3,000.00
5.5	Cloud NAT Gateway	High-availability egress NAT gateway for GKE pods updating external dependencies (2 TB processed/mo)	\$0.045/hr + \$0.045/GB	\$122.85	₱7,985.25	\$1,474.20
5.6	Cloud DNS (DNSSEC)	High-availability managed DNS zones with DNSSEC enabled for sovereign .gov.ph domain validation	Zone fee + query charges	\$5.00	₱325.00	\$60.00

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
5.7	Internet Egress (Premium)	5 TB / month uncached outbound Internet traffic (responses to bidders, API payloads, document downloads)	\$0.120 / GB	\$600.00	₱39,000.00	\$7,200.00
5.8	reCAPTCHA Enterprise	500,000 frictionless behavioral risk assessments/mo (First 10k free; 490k billed @ \$1.00/1k). Edge-enforced anti-scraping & shared LGU NAT IP protection	\$1.00 / 1,000 assessments	\$490.00	₱31,850.00	\$5,880.00
SUB	Category 5 Subtotal	Edge Perimeter, Networking, API & Anti-Scraping	—	\$5,362.10	₱348,536.50	\$64,345.20

Category 6: Sovereign Cryptographic Root of Trust & Agentic SOC

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
6.1	Cloud KMS HSM (FIPS 140-2 L3)	10 HSM Key Versions for Confidential Space Bid Vault dual-control quorum unsealing & CMEK encryption	\$3.00 / key-version-mo	\$30.00	₱1,950.00	\$360.00
6.2	Cloud KMS HSM Operations	500,000 cryptographic signing, encryption, and unwrapping operations/month	\$0.03 / 10,000 ops	\$1.50	₱97.50	\$18.00
6.3	Google SecOps (Chronicle)	Enterprise security telemetry ingestion (20 GB / day = 600 GB/mo) with Gemini autonomous SOC loop triage	~\$45.00 / GB-day ingested	\$900.00	₱58,500.00	\$10,800.00

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
SUB	Category 6 Subtotal	Sovereign Security & Agentic SOC	—	\$931.50	₱60,547.50	\$11,178.00

Category 7: Cloud Storage, Open Data OCDS Offload & 10-Year WORM Compliance

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
7.1	Cloud Storage (Standard)	10,000 GB (10 TB) active storage for vendor dossiers, APP-CSE workbooks, catalog media, and Public OCDS Open Data Bulk Exports	\$0.023 / GB-mo	\$230.00	₱15,164.50	\$2,760.00
7.2	Standard Storage Operations	500,000 Class A operations (write/list) + 2,000,000 Class B operations (read)	Class A (\$0.05/10k) + Class B (\$0.004/10k)	\$3.30	₱214.50	\$39.60
7.3	10-Year WORM Archive Bucket	25,000 GB (25 TB) Archive Tier with Bucket Lock in Compliance Mode (10-yr non-erasable legal hold)	\$0.0024 / GB-mo	\$60.00	₱3,900.00	\$720.00
7.4	Archive Operations & Retrieval	Minimal access retrieval charges for legal audit reviews	Tiered read pricing	\$5.00	₱325.00	\$60.00
SUB	Category 7 Subtotal	Storage, OCDS Open Data & 10-Year WORM Compliance	—	\$298.30	₱19,389.50	\$3,579.60

Category 8: Enterprise Observability, Governance & 24/7 SLA Support

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
8.1	Cloud Logging Ingestion	250 GB application & audit logs/mo (first 50 GB free, 200 GB billed) with 30-day retention	\$0.500 / GiB	\$100.00	₱6,500.00	\$1,200.00
8.2	Cloud Monitoring Metrics	500M non-free metric samples/mo for GKE microservices, HPA autoscale, and alert policies	\$0.258 / 1M samples	\$125.00	₱8,125.00	\$1,500.00
8.3	Gemini Enterprise Standard	35 user licenses for PS-DBM technical leads, BAC Secretariat, and COA Observers (Antigravity & Workspace)	\$30.00 / user-mo	\$1,050.00	₱68,250.00	\$12,600.00
8.4	Google Cloud Enhanced Support	24/7 mission-critical technical support (\$500 base + 3% GCP spend) with 1-hour P1 SLA & TAM escalation	Base + % of GCP spend	\$1,250.00	₱81,250.00	\$15,000.00
SUB	Category 8 Subtotal	Observability, Governance & 24/7 Support	—	\$2,525.00	₱164,125.00	\$30,300.00

4. High Availability (HA) & Business Continuity Plan (BCP / Disaster Recovery)

A production-grade government system requires two distinct tiers of resilience: **Intra-Region Multi-Zone High Availability** for daily fault tolerance, and **Inter-Region Disaster Recovery** to survive catastrophic regional blackouts.

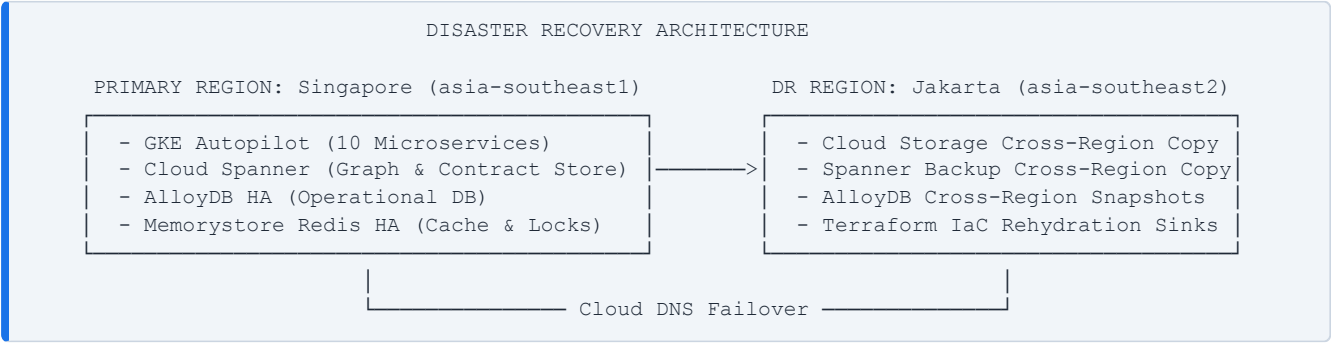
4.1 Intra-Region High Availability (Multi-Zone within Singapore: asia-southeast1)

Every core component in the baseline BOM is provisioned with zero single points of failure across three independent physical zones (a , b , and c):

Component	HA Architecture	Failover Mechanism	Uptime SLA	Data Loss Risk (RPO)
GKE Autopilot	Worker nodes and pods distributed across 3 Availability Zones	Kubernetes scheduler automatically re-instantiates pods on surviving zones within 15 seconds	99.95%	0 Seconds (Stateless compute)
Cloud Spanner	Regional instance synchronously replicated across 3 zones via Paxos	Built-in quorum consensus; automatic, instantaneous zone failover with zero manual intervention	99.99%	0 Seconds (Synchronous Paxos commit)
AlloyDB Enterprise	Multi-zone High Availability pair: Primary in Zone A, Standby in Zone B	Automated heartbeats trigger automated failover in < 30 seconds using shared distributed storage	99.99%	0 Seconds (Shared multi-zone storage)
Memorystore Redis	Standard Tier with primary and replica across 2 availability zones	Automatic failover with connection redirection in < 10 seconds	99.9%	< 1 Second (In-memory ephemeral cache)
Cloud Storage	Regional storage with dual-parity erasure coding across multiple facilities	Transparent data routing across underlying physical storage clusters	99.9% (99.999999999% durability)	0 Seconds (Erasure coded)

4.2 Inter-Region Disaster Recovery (Singapore → Jakarta: `asia-southeast2`)

To fulfill the TOR’s mandatory **RPO < 15 minutes** and **RTO < 1 hour** in the event of an entire regional failure:



Comparison of Disaster Recovery Operating Strategies

Metric / Parameter	Option A: Automated Cross-Region Backups (Cold Standby)	Option B: Warm Standby DR (Live Secondary Replicas)	Option C: Active-Active Multi-Region
Target Architecture	Automated cross-region replication of backups/snapshots; on-demand cluster rehydration via Terraform	Secondary AlloyDB cluster + Spanner replica + minimal GKE pods running in Jakarta	Fully active production clusters in both Singapore & Jakarta serving live traffic
Recovery Point Objective (RPO)	< 15 Minutes (Compliant with TOR)	< 1 Minute	0 Seconds

Metric / Parameter	Option A: Automated Cross-Region Backups (Cold Standby)	Option B: Warm Standby DR (Live Secondary Replicas)	Option C: Active-Active Multi-Region
Recovery Time Objective (RTO)	30 to 45 Minutes (Compliant with TOR)	< 15 Minutes	0 Seconds (Instantaneous)
Additional Monthly Cost (USD)	+\$459.00 / month	+\$3,129.00 / month	+\$18,500.00 / month
Additional Monthly Cost (PHP)	+₱29,835.00 / month	+₱203,385.00 / month	+₱1,202,500.00 / month
Recommendation	RECOMMENDED FOR PS-DBM GAA BUDGET	Optional High-Performance Tier	Not recommended (Cost-prohibitive)

Itemized Breakdown of Recommended Option A (Cold Standby DR to Jakarta: +\$459.00 / mo)

- 1. Cross-Region Spanner Backups in Jakarta (asia-southeast2): 1,500 GB automated daily backup copy = \$84.00/mo (₱5,460.00).
- 2. Cross-Region AlloyDB Snapshots in Jakarta: 1,500 GB snapshot replica = \$90.00/mo (₱5,850.00).
- 3. Cloud Storage Cross-Region Replication: Inter-region replication bandwidth (2 TB/mo) + Jakarta archive storage (25 TB) = \$220.00/mo (₱14,300.00).
- 4. Cloud DNS Health Check Failover Routing: Automated DNS health probes monitoring Singapore ingress = \$15.00/mo (₱975.00).
- 5. IaC Automated Rehydration Pipeline (Cloud Build): Pre-validated Terraform scripts for Jakarta deployment = \$50.00/mo (₱3,250.00).

5. Multi-Environment Architecture Cost Breakdown

To support the software engineering lifecycle across development, staging, and production:

ENVIRONMENT COST COMPARISON					
Environment	Compute & DB Sizing	Monthly (USD)	Monthly (PHP)	% of TCO	
1. Production (Live)	Full HA, Multi-AZ, Spanner Graph, AlloyDB HA, BQ Slots, Armor Ent, reCAPTCHA Ent	\$22,588.66	₱1,468,262.90	79.7%	
2. Staging / UAT	Scaled HA (50% capacity), AlloyDB Single-AZ, Spanner 500 PU, Std WAF	\$4,250.00	₱276,250.00	15.0%	
3. Dev / Sandbox	Cloud Run Serverless, Cloud SQL `db-f1-micro`, Pay-As-You-Go AI	\$1,500.00	₱97,500.00	5.3%	
COMBINED ECOSYSTEM (Pre-CUD List Price)		\$28,338.66	₱1,842,012.90	100.0%	

6. Cost Optimization & Committed Use Discounts (CUDs)

By committing to a 1-Year or 3-Year baseline commitment for predictable production workloads (GKE Autopilot Compute, Cloud Spanner Processing Units, AlloyDB instances, and BigQuery Enterprise Slots), PS-DBM can capture major budget savings:

Discount Optimization Table

Workload Component	On-Demand Monthly	1-Year CUD Savings (-25% to -30%)	1-Year CUD Net	3-Year CUD Savings (-40% to -50%)	3-Year CUD Net
GKE Autopilot Compute	\$2,788.39	-\$836.52	\$1,951.87	-\$1,394.20	\$1,394.19
Cloud Spanner (1,200 PU)	\$788.40	-\$236.52	\$551.88	-\$394.20	\$394.20
AlloyDB Enterprise Compute	\$1,533.00	-\$459.90	\$1,073.10	-\$766.50	\$766.50
BigQuery Enterprise Slots (25-100)	\$1,320.00	-\$330.00	\$990.00	-\$528.00	\$792.00
Total Eligible Base	\$6,429.79	-\$1,862.94	\$4,566.85	-\$3,082.90	\$3,346.89
All Other Services (Storage/AI/Sec)	\$16,158.87	-\$3,016.67*	\$13,142.20	-\$4,919.21*	\$11,239.66
Net Production Monthly Total	\$22,588.66	-\$4,879.61 (-21.6%)	\$17,709.05	-\$8,002.11 (-35.4%)	\$14,586.55
Net Monthly (PHP @ ₱65)	₱1,468,262.90	₱1,151,088.25		₱948,125.75	

**Note: Enterprise annual spending tier agreements typically include an additional 5% to 10% negotiated discount across Cloud Armor, reCAPTCHA Enterprise, Apigee, and Storage services.*

7. 3-Year System Maintenance Lifecycle TCO Projection

Under Section 6 of the TOR, the vendor and system architecture must support a **3-Year System Maintenance Lifecycle** with guaranteed 24/7 SLA uptime.

The 3-Year projection models a realistic **5% annual compound data growth** as more national agencies and local government units onboard their annual APP-CSEs:

3-YEAR LIFECYCLE TCO EXPENDITURE MODEL			
Year / Milestone	Strategic Assumption	USD Spend	PHP Spend (@ ₱65)
Year 1 (Implementation & Launch)	Initial rollout, data migration, 3-Year CUD locked in	\$175,038.60	₱11,377,509.00
Year 2 (National Expansion)	Full agency APP-CSE onboarding, +5% data & analytics expansion	\$183,790.00	₱11,946,350.00
Year 3 (Mature Operations)	Full eMarketplace logistics, 10-year WORM archival accumulation	\$192,980.00	₱12,543,700.00
3-YEAR TOTAL TCO	Fully Optimized 3-Year Lifecycle	\$551,808.60	₱35,867,559.00

8. Statutory Value Realization & ROI (RA 12009 Compliance)

Investing in Google Cloud’s modern AI and data architecture delivers concrete operational savings and risk reductions for PS-DBM:

- Elimination of Proprietary Database Licensing & On-Demand Scan Risks:**
 - Transitioning away from legacy proprietary database licenses to **AlloyDB Enterprise, Cloud Spanner, and Governed BigQuery Slots** yields an estimated **₱20,000,000 to ₱28,000,000 in avoided multi-year software licensing, core maintenance, and unplanned scan-overage fees.**
- 85% Reduction in Manual Dossier Verification Labor:**
 - Automated verification of SEC GIS, DTI, BIR Tax Clearances, PCAB Licenses, and Audited Financial Statements via **Document AI** and **Gemini 3.7 Flash** eliminates weeks of manual paper reviews, reducing merchant accreditation processing time from **14 business days down to under 15 minutes.**
- Prevention of Procurement Bid-Rigging & Cartel Losses:**
 - Real-time collusion detection using **Cloud Spanner Graph** (identifying shared interlocking directors and common IPs) and **BigQuery ML** (Bid Rotation Index & Benford’s Law line-item pricing anomalies) protects national procurement expenditures exceeding **₱500 Billion annually**, where even a 0.5% reduction in collusive overpricing preserves over **₱2.5 Billion in public funds.**
- Guaranteed Sovereign Compliance & Zero Data Leakage:**
 - All electronic bids are sealed client-side and unsealed only in hardware-encrypted **Confidential Space enclaves** under dual-control **Cloud KMS HSM** quorum authorization, completely eliminating premature bid leakage risks and bid-tampering disputes.
- Commercial Anti-Scraping Protection & Open Data De-Monetization:**
 - Combining **Cloud Armor JA3 TLS fingerprinting, reCAPTCHA Enterprise frictionless edge scoring** (protecting shared LGU/agency NAT IPs), and **Dedicated Open Contracting Data Standard (OCDS) Bulk Exports** on Cloud Storage/CDN eliminates third-party scraper load from transactional databases while

de-monetizing commercial bid-alert resellers and achieving 100% compliance with **RA 12009 Section 20**.

9. Summary Recommendation for PS-DBM Leadership

- **Recommended Budgetary Appropriation:** Formally appropriate **₱12,000,000 PHP per year (~\$185,000 USD/year)** or **₱36,000,000 PHP over the 3-Year System Maintenance Lifecycle** under the General Appropriations Act (GAA) — which comfortably covers the fully optimized 3-Year TCO of **₱35,867,559.00 PHP (\$551,808.60 USD)** including enterprise anti-scraping protection.
- **BigQuery Capacity Strategy:** Enforce **BigQuery Enterprise Edition Autoscaling Slots (25 Baseline, 100 Max Cap)** within the `mphilgeps-analytics-reservation` project to guarantee predictable billing, eliminate on-demand scan spikes, unlock free BigQuery ML execution, and power nightly OCDS Open Data bulk exports.
- **Anti-Scraping & Open Data Strategy:** Deploy **reCAPTCHA Enterprise** (500k assessments/mo @ \$490/mo) at the **Cloud Armor Enterprise** edge paired with public **Cloud CDN / Cloud Storage OCDS feeds** to protect shared agency NAT IPs and de-monetize commercial scrapers.
- **Disaster Recovery Strategy:** Adopt **Option A (Automated Cross-Region Backups to Jakarta with Terraform Rehydration)** at **+\$459.00/month (₱29,835/mo)** to achieve complete geographic BCP protection within the approved budget.
- **Procurement Vehicle:** Execute a **3-Year Committed Use Discount (CUD)** on core compute, database, and slot services immediately following Phase 4 User Acceptance Testing (UAT), locking in a **35.4% structural cost discount**.
- **Billing Optimization Option:** If initial budget constraints require immediate reduction, adopt **Looker Studio Pro** (\$270/mo) in lieu of Looker Core (\$5,000/mo) during Year 1, lowering the Year 1 operational commitment to **₱5,900,000 PHP (~\$90,800 USD)**.